

Vol. XI.

No. 4.

The Fern Bulletin.

A Quarterly Devoted to Ferns.

OCTOBER

Binghamton, N. Y.
THE FERN BULLETIN CO.
1903

Vol. XI.

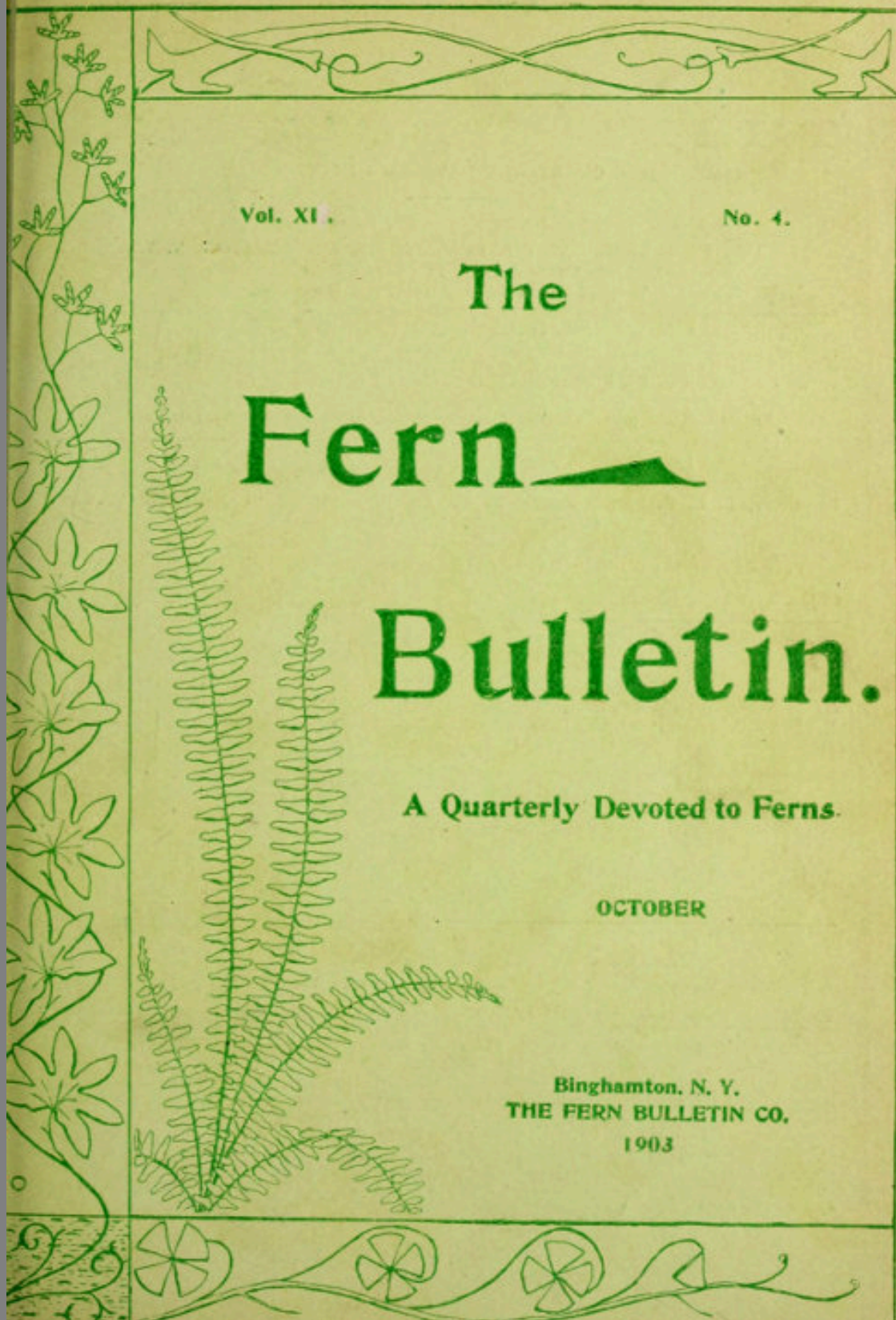
No. 4.

The Fern Bulletin.

A Quarterly Devoted to Ferns.

OCTOBER

Binghamton, N. Y.
THE FERN BULLETIN CO.
1903



The Project Gutenberg eBook of The Fern Bulletin, October 1903

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Fern Bulletin, October 1903

Author: Various

Editor: Willard Nelson Clute

Release date: April 11, 2018 [eBook #56959]

Language: English

Other information and formats: www.gutenberg.org/ebooks/56959

Credits: Produced by Stephen Hutcheson and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK THE FERN
BULLETIN, OCTOBER 1903 ***

The Fern Bulletin.

Vol. XI. No. 4.

A Quarterly Devoted to Ferns.

OCTOBER

Binghamton, N. Y.
THE FERN BULLETIN CO.
1903

THE FERN BULLETIN

A QUARTERLY DEVOTED TO FERNS

WILLARD N. CLUTE, Editor

THE FERN BULLETIN CO., PUBLISHERS, BINGHAMTON, N. Y.

20 CENTS A COPY; 75 CENTS A YEAR.

Awarded Grand Prize at the Paris Exposition.

To insure subscribers against loss of one or more numbers between the expiration and renewal of their subscriptions the journal will be sent until ordered stopped. All arrearages must be paid. PERSONAL CHECKS MUST CONTAIN TEN CENTS EXTRA FOR COLLECTION. Otherwise credit will be given for the amount less collection fees.

Entered at the postoffice, Binghamton, N. Y., as second-class mail matter.

THE LINNAEAN FERN CHAPTER

PRESIDENT, B. D. GILBERT, Clayville, N. Y. SECRETARY, HOMER D. HOUSE, N. Y. Bot. Garden, Bronx, New York City.

Fern students are cordially invited to join the Chapter. Address either the President or Secretary for further information. The annual dues are \$1.00 and should be sent direct to Jas. A. Graves, Treasurer, Susquehanna, Pa.

FERNS FOR SALE

A Fern Student of many years standing who has made a specialty of cultivating New England Ferns is prepared to supply plants for Ferneries and House. For prices and variety address,

C. C. BROWNE, SOUTH GROVELAND, MASS.

“MOSESSES WITH A HAND LENS”

BY DR A. J. GROUT

It is the only book of its kind in the English language. It makes the mosses as easy to study as the flowering plants. Eight full page plates and ninety figures in the text. Price \$1.10 postpaid. Send for sample pages to O. T. Louis, 59 Fifth Avenue, N. Y. City.

WANTS AND EXCHANGES

Special announcements inserted here for One Cent a word. No notice received for less than 25c. No charge for address.

EXCHANGE—I will exchange three flowering plants of California for any one desired fern of the United States. Send me your list of duplicates. GEORGE B. GRANT, 637 Summit Ave., Pasadena, California.

EATON'S FERNS FOR \$35.00

Eaton's "Ferns of North America" has been out of print for some time and is constantly advancing in price. We can offer a second-hand copy, the two volumes bound in cloth, clean and in good condition, for \$35.00, express paid. There are 81 colored plates, and all the North American ferns are described. Address,

THE FERN BULLETIN, Binghamton, N. Y.



WILLIAM RALPH MAXON.

CONTENTS

<u>THE FERN FLORA OF NEW YORK.</u>	97
<u>Ophioglossaceae.</u>	98
<u>Osmundaceae.</u>	99
<u>Schizaeaceae.</u>	100
<u>Polypodiaceae.</u>	100
<u>Equisetaceae.</u>	103
<u>Isoetaceae.</u>	103
<u>Lycopodiaceae.</u>	103
<u>Salviniaceae.</u>	104
<u>Selaginellaceae.</u>	104
<u>FERNWORT NOTES—IV.</u>	105
<u>SCOLOPENDRIUM FROM CANADA.</u>	107
<u>THE GENUS EQUISETUM IN NORTH AMERICA.</u>	108
<u>THE SPECIES-CONCEPTION AMONG THE TERNATE</u> <u>BOTRYCHIUMS.</u>	115
<u>NEW FORMS OF FERNS.</u>	118
<u>FERNS IN BOTTLES.</u>	120
<u>WILLIAM RALPH MAXON.</u>	121
<u>ANOTHER STATION FOR ASPLENIUM EBENEUM HORTONAE.</u>	122
<u>INDEX TO CURRENT LITERATURE RELATING TO FERNS.</u>	122
<u>EDITORIAL.</u>	123
<u>BOOK NEWS.</u>	125
<u>A WORD FROM THE EDITOR</u>	129

VOL. XI.

OCTOBER, 1903.

No. 4

THE FERN FLORA OF NEW YORK.

BY B. D. GILBERT.

The State of New York has the largest area of any northern State east of Michigan. It also possesses a great diversity of surface, with its two mountain ranges, its numerous lakes, its interior salt basin, and its seashore confined entirely to the southern extremity. On its eastern side it stretches through more than four degrees of north latitude, and as these are the degrees just south of the 45th parallel, it is easy to understand that there is liable to be a greater intermixture of northern and southern forms of ferns than there would be in a State lying farther south. And the fact is that certain species from the north and others from the south do meet within its borders. This also accounts for the large number of species found in the State: California and Texas, the one State having four times the area of New York, and the other five times that area, being the only ones which contain as large or a larger number of species.

For the purpose of fern classification, the State may be divided into four distinct zones, as follows:

I. THE LITTORAL.—This comprises Long Island and Staten Island. Only one fern is peculiar to this zone, viz. *Woodwardia angustifolia*; but there are two Lycopods, viz. *L. alopecuroides* and its variety *adpressum*.

II. THE CATSKILL MOUNTAIN REGION, extending down to Manhattan Island.—This being the southern mountain range of the State, it is here that three southern species find their northern limit, viz. *Asplenium Bradleyi*, *A. montanum* and *Cheilanthes vestita*. It may be a question whether the Connecticut stations for *Asplenium montanum* lie farther north than the New York stations, but it is certain that there can be but little difference between them in this respect.

III. THE ADIRONDACK REGION, extending as far south as Little Falls.—Here there are a few of the northern species that descend to their southern limit in

this country. Among them may be mentioned *Nephrodium fragrans*, *Polystichum Braunii*, *Woodsia glabella*, and *W. hyperborea*. There are also two Lycopods to be included in this list, *L. annotinum pungens* and *L. Sitchense*. 98

IV. THE WESTERN REGION, extending from the mountain regions to the State's western boundary, the southern part drained by the Susquehanna and its tributaries, and the northern part containing (a) *The Salt Basin* of Syracuse and its vicinity, the home of *Scolopendrium* and *Botrychium Onondagense*, and (b) *The Central Basin*, extending from Little Falls to Oneida Lake, and drained by the Mohawk River, being famous for its large number of *Botrychia*, some of which seem to be almost, if not quite, peculiar to this region.

In the preparation of this list I have consulted various local floras and other reports of the State, especially Paine's "Catalogue of Plants Found in Oneida County and Vicinity;" the Annual Reports of the State Botanist, "The Flora of the Upper Susquehanna," and the files of the FERN BULLETIN. I have also been favored with many notes from the State Herbarium sheets by Professor Peck himself; while my own herbarium and that of Dr. J. V. Haberer, of Utica, have been exceedingly useful in fixing definite localities.

The plan I have followed is to include along with the species only such varieties as seem to be most important and distinct.

Mr. Bush in *The Torrey Bulletin* for June, 1903, enumerates 59 species and varieties of ferns found in Texas. This compares well with our list of 64 species and varieties in New York State, of which only 18 of these are common to both States. Mr. Reverchon's list for Texas, published in the FERN BULLETIN, April, 1903, gives 51 ferns and 15 allies, or 66 in all. The present list includes 53 species and 12 varieties of ferns, and 23 species and 7 varieties of fern allies. If we take Bush's ferns and Reverchon's allies, there are 74 species and varieties known in Texas, or 20 less than I have enumerated in New York.

OPHIOGLOSSACEAE.

Botrychium lanceolatum Angs. Open upland woods in the eastern part of State. Not reported from western part. Scarce.

Botrychium matricariaefolium A. Br. In same localities with *B. lanceolatum*, but more plentiful.

Botrychium obliquum Muhl. GRAPE FERN. Abundant in pastures and woodlands, especially in the central part of the State. 99

Botrychium obliquum dissectum Clute. In similar places with *B. obliquum* and always associated with it. Frequent.

Botrychium obliquum f. elongatum Gilbert & Habereri. Scarce in pastures, eastern New York.

Botrychium obliquum Habereri Gilbert. Scarce. Mohawk Valley, Catskill Mts., and probably occasional elsewhere.

Botrychium obliquum intermedium Clute. Pastures. Occasional in the eastern part of the State.

Botrychium obliquum matricarioides Gilbert. Beaver Lake. Lewis Co.; Whitestown: Gilbert. I have seen no other specimens from this State, but Dr. Underwood cites "Northern New England and New York." (*Osmunda matricariae* Schrank.)

Botrychium obliquum Oneidense Clute. Pastures and open woods. Occasional.

Botrychium obliquum tenuifolium Gilbert. Field near top of mountain. Shandaken, Ulster Co., Mary F. Miller. As I possess tracings of Dr. Underwood's type specimens of *B. tenuifolium* Und., I am able to say that these specimens from Shandaken match the tracings completely. (*B. tenuifolium* Und.)

Botrychium Onondagense Underwood. Within a radius of five miles from Syracuse. The stations there are the only ones known in the State.

Botrychium simplex E. Hitchcock. Damp meadows and sandy swamps in Central and Northern New York. Also Wading River. L. I., *E. F. Miller*.

Botrychium tenebrosum A. A. Eaton. Baldwinsville, *L. M. Underwood*; Deerfield and Cedar Lake. *J. V. Haberer*. Probably not rare, but escapes notice because of its minute size.

Botrychium Virginianum Sw. RATTLESNAKE FERN. In rich woods. Common.

Ophioglossum vulgatum L. Frequent in damp ground throughout the State.

OSMUNDACEAE.

Osmunda cinnamomea L. CINNAMON FERN. Swamps. Common.

Osmunda Claytoniana L. INTERRUPTED FERN. Along roadsides and in thickets. 100

Osmunda regalis L. ROYAL FERN. Widely spread, but not so abundant as *O. Cinnamomea*.

SCHIZAEACEAE.

Lygodium palmatum Swz. CLIMBING FERN. Hunter, Green Co. and McDonough, Chenango Co., *Professor Peck*. The only localities known in the State.

POLYPODIACEAE.

Adiantum pedatum L. MAIDEN HAIR. Plentiful throughout the State.

Asplenium angustifolium Michx. NARROW-LEAVED SPLEENWORT. Damp, shady soil, common.

Asplenium Bradleyi D. C. Eaton. Near Newburg, *D. C. Eaton*; Shawangunk Mts., *C. Lown* in State Herbarium. Rare.

Asplenium ebeneum Ait. EBONY SPLEENWORT. Frequent on edges of woods or rocks throughout.

Asplenium ebenoides R. R. Scott. Near Saugerties, Ulster Co., C. Lown, in State Herbarium. Rare.

Asplenium montanum Willd. On rocks about Lakes Mahonk and Minnewaska, Prof. C. H. Peck. New Paltz, H. Denslow. Rare.

Asplenium ruta-muraria L. WALL RUE SPLEENWORT. Limestone cliffs. Helderberg Mts., Professor Peck. Spraker's and Chittenango Creek, Paine. Little Falls, Gilbert. Not common.

Asplenium Trichomanes L. MAIDEN HAIR SPLEENWORT. On limestone rocks. Middle and eastern parts of the State. Localities comparatively few. Common in the southern part, Clute.

Athyrium filix-foemina Roth. LADY FERN. Very common everywhere. About 15 varieties occur in State.

Athyrium Thelypteroides Desv. SILVERY SPLEENWORT. Damp woods. Common.

Camptosorus rhizophyllus Link. WALKING FERN. On shaded rocks. Not common but widely scattered.

Cheilanthes vestita Swz. Rare. Washington Heights, Manhattan Island, W. W. Denslow in herb. Gilbert; Poughkeepsie, Professor Peck.

Cystopteris bulbifera Bernh. Rocky banks and ravine sides near water. Common in the central, rare in the southern part of the State.

101

Cystopteris fragilis Bernh. FRAGILE BLADDER FERN. On gravelly hillsides and moist rocks. Common. Two varieties of this are occasionally met with, viz. *dentata* Hook, and *magnasora* Clute.

Dicksonia pilosiuscula Willd. Abundant. Growing in large beds along roadsides and in moist woodlands.

Nephrodium Boottii Davenp. Frequent in damp woods throughout the State.

Nephrodium cristatum Michx. CRESTED FERN. Common in swampy grounds.

Nephrodium cristatum Clintonianum Gilbert. Growing generally with the type.

Nephrodium fragrans Rich. FRAGRANT FERN. Lake Avalanche and Cascadeville on cliffs. *Professor Peck*. Rare.

Nephrodium Goldienum Hook. Occasional in swampy ground throughout the State.

Nephrodium marginale Michx. Plentiful in rocky woods.

Nephrodium Noveboracense Desv. NEW YORK FERN. Common in damp woods and thickets.

Nephrodium simulatum Davenp. Middle Village, L. I., *Rev. G. D. Hulst*; Babylon and Bellville, L. I., *W. N. Clute*; Oneida Lake, *H. D. House*. Rare.

Nephrodium spinulosum Desv. Probably rather common, but not recognized. Grows in damp woods throughout the State.

Nephrodium spinulosum dilatatum Baker. Infrequent. Catskill and Adirondack Mts., *Professor Peck*; Yates Co., *Sartwell*. In more elevated situations than the type.

Nephrodium spinulosum f. intermedium. Davenp. Our commonest woods fern.

Nephrodium thelypteris Desv. MARSH FERN. Abundant in wet ground, whether shaded or not.

Onoclea sensibilis L. SENSITIVE FERN. Common in swampy ground.

Pellaea atropurpurea Link. On cliffs both in eastern and western New York. Scarce.

Pellaea gracilis Hook. SLENDER CLIFF BRAKE. Not common, but occurring in many localities through the State. Always on rocks or cliffs.

Phegopteris Dryopteris Fee. Common in rich, damp woods.

Phegopteris hexagonoptera Fee. Rather scarce, but more frequent in the southern and western parts of the State.

Phegopteris polypodioides Fee. Common on wet rocks and in damp woods.

Polypodium vulgare L. COMMON POLYPODY. Common on rocks.

Polypodium vulgare cristatum Lowe. Rock City, Dutchess Co., *Charles A. Coons*.

Polystichum acrostichoides Schott. CHRISTMAS FERN. Common in woods. The variety *incisum* with the type.

Polystichum Braunii Lawson. Summit, Schoharie Co., Catskill and Adirondack Mts., *Professor Peck*; Ilion ravine, *Rev. H. M. Simmons* in herb. *Gilbert*.

Pteris aquilina L. BRACKEN. Common throughout the State.

Pteris aquilina pseudocaudata Clute. The common form on many parts of Long Island. *Clute*.

Scolopendrium vulgare J. E. Sm. HART'S TONGUE. Rare. Only in ravine of Chittenango Creek, Green Lake, Jamesville and a few other stations in the vicinity of Syracuse.

Struthiopteris Germanica Willd. OSTRICH FERN. Frequent in the Valley of the Mohawk and its tributaries. Western part of the State, *Torrey*; Southern tier, *Clute*.

Woodsia glabella R. Br. Crevices of rocky ledges at Lake Avalanche and in the pass north of it, *Professor Peck*; Haines' Falls, Catskill Mts., *Professor Peck*. The station at Little Falls has been destroyed by excavation.

Woodsia hyperborea R. Br. Rare. Adirondack Mts. Only three stations known—at Cascadeville, Lake Avalanche, and Ampersand Mt., *Professor Peck*.

Woodsia Ilvensis R. Br. RUSTY WOODSIA. On rocks and cliffs. Rather scarce.

Woodsia obtusa Torr. In rich woods and on rocks. Widely spread but not common.

Woodwardia angustifolia Sm. Flatbush and Middle Village, L. I., *Professor Peck*; Babylon and Bellville, L. I., *Clute*; Staten Island, *Torrey*.

Woodwardia Virginica Sm. CHAIN FERN. Frequent in swamps from Long Island to the western part of the State. 103

EQUISETACEAE.

Equisetum arvense L. FIELD HORSETAIL. Very common in either wet or dry soil. Mr. Eaton describes a dozen different forms.

Equisetum fluviatile L. Borders of lakes and ponds. Not uncommon.

Equisetum palustre L. Swamps near Buffalo, *Clinton*, according to Eaton. Rare.

Equisetum hiemale L. SCOURING RUSH. Very common in swampy and gravelly places. Eaton describes seven forms.

Equisetum hiemale intermedium A. A. Eaton. Oneida Lake, *Dr. J. V. Haberer*. Rare.

Equisetum scirpoides Michx. Shaded ravines and sphagnum marshes. Widely scattered but scarce.

Equisetum sylvaticum L. WOOD HORSETAIL. Marshes and borders of woods. Not uncommon. Eaton enumerates seven forms.

Equisetum variegatum Schleich. Brisbin Swamp, Coville; Oriskany and shores of Lake Ontario, Paine; Owasco Lake, Prof. I. H. Hall; along W. Canada Creek, near Herkimer, J. V. Haberer.

ISOETACEAE.

Isoetes Amesii A. A. Eaton, sp. n. (*I. riparia Canadensis* Engelm. Also *I. Dodgei* Eaton.) Peekskill is the only known locality in this State, but the species is very common in New England.

Isoetes echinospora Braunii Engelm. The common form in New York State.

Isoetes echinospora muricata Engelm. Lake Luzern, A. A. Eaton. Rare.

Isoetes Engelmanni A. Br. Abundant along the Susquehanna river in Broome and Tioga counties, Clute. New York City, Buchheister.

Isoetes macrospora Durien. Catskill Mts., Schweinitz, according to A. A. Eaton. (*I. lacustris pauperula* Engelm.) Rare.

LYCOPODIACEAE.

Lycopodium alopecuriodes L. Babylon, L. I., Clute. Rare.

Lycopodium alopecuroides adpressum Chapm. Babylon, L. I., Clute; Forbell's Landing, L. I., M. Zimmerman. Rare. 104

Lycopodium alopecuroides f. *polyclavatum* McDonald. Staten Island, Clute. Rare.

Lycopodium annotinum L. Damp woods. Rather scarce, especially in western part of State.

Lycopodium annotinum pungens Spring. Summits of Mt. Marcy and Mt. McIntyre, Professor Peck.

Lycopodium clavatum L. COMMON CLUB MOSS. Abundant in open woods and thickets. The var. *monastachyon* Hook. was found on Bald Mt., Fulton Chain, by Dr. Haberer.

Lycopodium complanatum flabelliforme, Fernald. Common.

Lycopodium complanatum Wibbei var. n. J. V. Haberer. Scarce. Mohawk Valley. Haberer in herb. Gilbert and in herb. Gray, No. 1,953.

Lycopodium inundatum L. Bogs and borders of swamps. Frequent except in southern and western part where it is rare.

Lycopodium inundatum Bigelovii Tuckerm. Riverhead, L. I., *Professor Peck*.

Lycopodium lucidulum Michx. On damp grounds or rocks. Common.

Lycopodium obscurum L. Woods. Common.

Lycopodium Selago L. Summits of Mts. Marcy and McIntyre and in Indian Pass. *Professor Peck*. Alpine and rare.

Lycopodium Sitchense Rupr. In pass between Nipple Top and Mt. Colvin and on side of Mt. Marcy, *Professor Peck*. Alpine and rare.

Lycopodium tristachyum Pursh. Cold Spring Harbor, L. I., *Prof. F. E. Lloyd*; Ray Brook, Essex Co., Caroga, Fulton Co., *Professor Peck*. Alden Creek, *Gilbert*. Infrequent.

SALVINIACEAE.

Azolla caroliniana Willd. Floating on water. Frequent. Cayuga Lake, Black Creek, *Professor Peck*; in all side waters of Lake Ontario, *Paine*.

SELAGINELLACEAE.

Selaginella apus Spring. Wet banks and pastures. Uncommon. Cedar Lake, Herk. Co., *Paine*; borders of Rome swamps, *Kneiskern*; head of Oneida Lake, *Haberer*; Unadilla, *Brown*.

Selaginella rupestris Spring. On dry open rocks. Rather scarce. Little Falls, *Paine*; Adirondacks. *Professor Peck*; Washington Heights, *W. W. Denslow*. Plentiful in the extreme southeast, *Clute*.

FERNWORT NOTES—IV.

BY WILLARD N. CLUTE.

NEPHRODIUM MOLLE IN FLORIDA.—Mr. James H. Ferriss recently called my attention to specimens of *Nephrodium molle* received from Florida with the suggestion that this species might be native to the State. On this point, Reasoner Brothers, the well-known plant dealers, have written him that they no longer grow the fern since it is abundant in a wild state and easily obtained when wanted. There seems to be no reason why the species should not occur in Florida, since *Nephrodium patens*, a close ally, is common there; but as *N. Molle* is not listed from the United States, we publish this note in the hope of drawing out further information about it and of ascertaining if possible whether *Molle* is actually native, or only a well naturalized escape. Superficially, *molle* and *patens* are so very much alike that it is very easy to confuse them. The venation, however, is a sufficiently distinct feature. In *patens* the basal veins in each pinnule run to the sinus, uniting at, or just below, it; in *molle* they unite at some distance from the sinus from whence a single vein runs to the sinus. According to Jenman, *patens* has a creeping horizontal rootstock with the fronds arranged in two lines along it, while *molle* has an erect rootstock. The fronds of the latter are also softer and thinner.

NATURALIZATION OF AN EXOTIC FERN.—Records of ferns becoming naturalized in new regions are very rare. *Pteris serrulata* is probably our most conspicuous American example, having been found as an escape in several places, while it is known to grow abundantly on old walls in New Orleans. I have also reported the occurrence in the same place, of an abundance of *Pteris longifolia* previously known in the United States from Florida alone. In the FERN BULLETIN for January, 1898, mention is made of fronds of *Pteris tremula* seventeen inches high collected from the walls of a tunnel in New York City, and the same article mentions a Japanese species of *Athyrium* that has become naturalized on Staten Island, New York. To this meagre list, it is with much pleasure that I add another species in the shape of the Japanese climbing fern (*Lygodium Japonicum*). This

Mrs. A. P. Taylor has sent to me from Thomasville, Georgia, where she finds it in profusion along the sides of a deep ditch. The station is not far from a greenhouse from whence the plants doubtless came in the first place, but all indications point to a further spread of this pretty and interesting species.

THE FORMS OF THE SPINULOSE WOOD FERN.—It is well-known to fern students that much more attention has been paid to the forms of ferns on the other side of the Atlantic than on this. Since the same species are often common to both localities, it is but natural that the early students of American ferns should pay rather more attention to the mere forms of species common to Great Britain and America than their systematic importance warrants. This is especially true of the variable *Nephrodium spinulosum* whose variety *intermedium*, I am convinced, is scarcely more than an ecological form. In this view I am glad to be borne out by Mr. A. B. Klugh, who has recently examined nearly 500 Canadian specimens and come to the same conclusion. Mr. Klugh writes: “In number of glands on the indusium, in color of scales on the stipe, in shape and cutting of the frond and in degree of obliquity of the pinnae, we have a perfect gradation from true *spinulosum* to typical *intermedium*. Our commonest form has the indusium glandular and the scales of the stipe pale brown without a dark centre.” In a series of fronds examined there seemed to be no correlation between the color of the scales and the glands on the indusium, there being fronds with light scales and no glands, others with dark centered scales and many glands, and still others the exact opposites of these. *Intermedium* may be distinguished as a form, but it is certainly far less distinct than such plants as *Nephrodium cristatum* *Clintonianum* or *Pteris aquilina pseudocaudata* and would probably never have appeared in our lists but for the fact that much has been made of the forms of this species in other lands.

ELEVATION AND LYCOPODIUM SELAGO.—Some time ago I noted in this series, that a party of botanists on a visit to Mt. Ktaadn had found *Lycopodium selago* grading into *L. lucidulum* as they traveled downward from the summit, and quoted their opinion that *L. selago* is a xerophytic form of *L. lucidulum*. In regard to this, Mr. J. B. Flett writes that if the one intergrades with the other, it is doubtless due to elevation or cold, and not to xerophytic conditions. As to the plant’s habitat in the northwest, he says: “I

have never seen *L. selago* growing in a really dry place, I have studied this form in the field from Washington through British Columbia into the islands of southwestern Alaska and on the Aleutian Islands, also on the tundra between Cape Nome and Cape York. No one familiar with this tundra region would ever assert that there are any xerotic forms on it.”

SCOLOPENDRIUM FROM CANADA.

BY HOMER D. HOUSE.

At least four stations for the Harts-tongue fern are known in the vicinity of Owen Sound in northwestern Ontario. Specimens from these localities are rare in herbaria, and the writer is fortunate in receiving specimens from near Collingwood, a station twenty-three miles east of Owen Sound. This station was first authentically reported by Mr. Osler and described by Mr. Maxon in "Fernwort Papers" in 1900. These specimens were collected by Dr. W. A. Bastedo and he describes the place where they were collected as being five or six miles from Collingwood. The plants were growing in a shady, though rather open wood, along the course of a small stream. The altitude is given as 1635 feet above sea-level. The plants at the time of collection, July 17th, 1903, were nearly all young and even the mature fronds are but five to eight inches in length, though all of them are very broad for their length. Dr. Bastedo further notes that in the recesses of the cliff, snow was still abundant at that date. *Polystichum Lonchitis* and *Dryopteris Filix-mas* were abundant and *Asplenium Trichomanes* and *Cryptogramma Stelleri* were common upon the cliffs. This station is undoubtedly one of those described by Mr. Maxon in the neighborhood of Collingwood. However, a careful search of this entire region is very much to be desired, as it is probable that the fern has a more general distribution in this region than is known at present.

THE GENUS EQUISETUM IN NORTH AMERICA.

BY A. A. EATON.

FIFTEENTH PAPER. VARIETIES OF E. HIEMALE.

1. *Intermedium* A. A. Eaton. Stems 1 to 4 feet high, 1 to 4 lines in diameter, simple or ultimately branched, 20 to 30 angled, rough with transverse bands of silex or becoming smoother by a later deposit covering them; sheaths longer than broad, amplified, green excepting narrow black and white incurved limb, or exceptionally with other black and white markings; leaves keeled below the middle, flat and often centrally grooved above; teeth thin, brown, hyaline-bordered, deciduous or persistent; anatomy of *hiemale* as previously described. New York, Michigan and westward. Common west of the Mississippi, being an important forage crop in some States. The anomalous *laevigatum* collected by Rydberg at Thedford, Neb., No. 1283 (Cont. Nat. Herb. III, 194), is this variety, as is the plant referred to under the name of *variegatum* by V. K. Chestnut (Cont. Nat. Herb. VII, 304), as used for various unimportant purposes by the Indians of Mendocino County, California. He also mentions the fact that horses eat it even when grass is abundant.

2. *Texanum* Milde. Stems erect, very slender, somewhat rough, 10 to 12 angled, hardly 1 foot high, dirty green; sheaths elongated, slightly widened, 2 to 2 1-3 lines long and 1 1-3 wide, concolorous, leaves flat, centrally grooved and 4 angled above and centrally ridged below; teeth persistent, flexuous, white with red-brown center, lance subulate, smooth, only the lowermost three sheaths red-brown; ridges convex; carinal bast 7, vallecular 4, cells high, vallecular holes transverse oval; stomata rows separated by 7 to 8 cells, grooves naked, lumen of epidermal cells very wide, angles with broad, short bands, never with two rows of tubercles. Texas, *Chas. Wright*.

This is Milde's description. I have never seen this plant. Milde states that it is a very peculiar plant that equals the weakest specimens of var. *Moorei*,

but differs greatly from it, and he asks if it may not be the young stage of a larger species.

3. *Herbaceum* var. nov. Cespitose, decumbent, ascending or erect, 3 to 10 inches high, $\frac{1}{2}$ to 1 line in diameter, 6 to 12 angled, weak and herbaceous or becoming firmer the second year, usually bearing a single branch 1 to 2 inches long at each node. Walls of the stem thicker than in *hiemale*; ridges with long cross-bands; grooves naked, except for small spots of silex on the cells; sheaths elongated and very wide-spreading, with a narrow black band at tip, otherwise green or (in dried specimens at least) all suffused with black; leaves 3-angled or flat in the middle above, rarely bearing a central groove; teeth fuscous, flexuous, deciduous, leaving a hard, horny, centrally grooved erect or incurved, usually shining, borderless leaf base $\frac{1}{2}$ its height; spikes narrowly elliptical, rounded, not apiculate. *Coville & Funston*, 1297, Death Valley Exp., banks of Kaweah river at Three Rivers, Tulare Co., Calif., July 26, 1891 (Nat. Herb., 25, 101), as *variegatum*. Three little plants, 3 inches high, well fruited (Cont. Nat. Mus. IV, 226). *C. & F.*, 1042, 1 mile south of Kernville, Kern Co., Calif., on north fork of Kern river, Alt., 750 meters, June 23, 1901, as *variegatum* (Nat. Herb., 25100).

109

In some of its characters, such as sheaths and persistent, incurved leaf-bases, this plant resembles *Funstoni*, but the section is similar to *hiemale*. An abundance of material might show this to be a good species. The only thing I have seen that approaches it in texture is *E. Sieboldi* Milde from Japan, which is even more grass-like.

4. *Pumilum* var. nov. Cespitose; stems in a dense cluster, 6 to 15 inches tall, 8 to 16 angled, $\frac{1}{2}$ to 1 line in diameter, mostly geniculate at the lower nodes, nearly all the joints tumid, the lower gibbous; ridges with cross-bands of silex, grooves naked; sheaths tight, often symmetrical through the tumidity of the node, narrowest in the middle except where nodes are normal, bearing a broad black band below and a narrower black limb, the two separated by a pinkish or dirty white band which is often suffused with black or even entirely black towards the top of the stem, fading to dirty ashy the second year, ultimately splitting, recurving and falling off in patches; leaves linear, erect, prominently 3-angled, the central one sometimes

grooved on the smallest stems and branches; teeth persistent, dark brown, somewhat flexuous, white-bordered for 1-5 to 1-4 their height.

110

Found at intervals for a mile along the railroad grade at North Hampton, N. H. At the foot of the grade, in moist soil near a brook, probably from the same source as this, a form of *affine* grows, but the joints are often tumid and occasionally geniculate, the branches when present like stems of this, tumid jointed, often so gibbous as to rupture the sheath. Peculiar for its small cespitose stems, dark sheaths and especially the tumid or gibbous nodes, which make the stems thickest there, while usually the nodes are contracted.

This is near the European variety *viride* Milde, but differs in having bands on the ridges, no rosulæ in the grooves, and in the tumid joints.

5. *Suksdorfi* var. nov. Stems 1 to 2½ feet high, 1 to 3 lines wide, about 24 angled, rough, with cross-walls of silex, rarely with ends elevated to two rows of tubercles; stomata in single rows, rarely double for a short distance, each stoma connected at top and bottom with its opposite by rows of rosulæ formed by the silex bands of the grooves throwing up tubercles on each cell of the epidermis, which open at top to circular jagged disks, these often obscured later by a washing of silex, but always shown near the tops of the stems and on the branches; sheaths elongated, cylindrical, tight, black, developing a ring of tawny white which gradually increases till it occupies the whole sheath except a narrow black basal ring and a narrow black limb formed by the horny tips of the leaves; leaves linear, narrowed above the middle, the lower 2-3 keeled, the upper third flat, rarely with a narrow carinal groove above, tipped with a small, black, horny, hyaline-bordered point; teeth articulated to the leaves, black-centered, soon fading, withering and deciduous.

Anatomy of *hiemale*, the carinal bast elongated along the dissepiment, the vallecular much smaller but often similar in shape. Upper 1 to 3 nodes bearing 1 to 4 branches each, which overtop the stem and bear contemporaneous spikelets.

This would be a noteworthy variety even if it bore no branches. It is the only American form of *hiemale* known to me, except occasionally *intermedium* which bears branches with the first effort of growth. All 111 the others develop them, if at all, after the stem has ceased to grow, and the vegetative energy, having no other outlet, pushes out a few of the latent buds lying between the ridges at the nodes.

Bingen, Wash. High bottom land on the Columbia river. W. N. Suksdorf, September 3, 1902, No. 2161.

6. *Drummondi* (Milde) *C. robustum Drummondi* Milde, Mon. Equis. 593. Fertile stems 3 feet high, 16 angled; sheaths short, the lowest fuscous; teeth persistent, white, crispate; stomata often of 1 to 3 lines to a series, which are separated by 4 to 6 cells.

Collected by Drummond at the Brazos river in Texas. It is very aberrant, but is placed here on account of its anatomy. I have not seen specimens of this.

7. *Affine* (Eng.) (*E. robustum affine* Eng.) *E. hiemale* of American authors, not L. Stems 18 to 30 inches high, 2 to 5 lines in diameter, finely 16 to 40 angled, dark green, angles with broad bands of silex, rarely with two rows of tubercles. Internodes when dry contracted above and below, widest in the middle as in *hiemale*, scurfy when young; sheaths longer than broad, at first with a black limb, developing a broad ashy band and narrow black basal ring, fading, rupturing and deciduous the second or third year; leaves narrowly linear, sharply 3 angled, the central ridges only rarely centrally grooved except on the branches, where they usually are; commissural groove very narrow, not widened upward; teeth articulated to the sheaths, persistent or usually cohering by their tips and torn off by the growth of the stem, those of each sheath shaped like a candle extinguisher, all telescoped together and borne up on the tip of the stem.

Very common in New England and the east generally, where the type of *robustum* is absent. Toward the west it runs into the next, but it is occasionally found, even to the valley of Mexico (Pringle 3329). Approaches typical *hiemale* in its long sheaths and size, and differs little except in the cross-bands of silex. Found usually in moist sand near a watercourse; at times on high sandy banks. It is by no means certain that

this is the variety described by Engelmann under this name, but from the brief description he gives it seems safe to assume that it is. Two branched forms are found, as follows:

a. *Forma ramosum* f. nov. (f. *Ramigerum* A. A. E., in Gilbert's list, p. 26, not A. Br. in Sched., which normally branches at the 3 to 5 middle nodes.) Stems issuing one to several branches from the upper nodes after the death of the top of the main axis; teeth usually persistent and leaves centrally grooved. b. *Forma polystachyum* Prager. Stems issuing small spiciferous branches late in the season. As remarked by Mr. Gilbert (List, p. 26), these forms are seldom found together and many patches show neither.

The stems of this variety persist at least three years and probably longer. I have found but two causes of death, old age not appearing as a factor. Both are fungoid. After the stem has persisted for a time small white patches appear under the epidermis of the upper internode. These increase in number and the internode finally dies, not, however, till the second one shows the disease. This may continue till the whole stem succumbs. The other fungus is a smut that breaks out in small pustules, finally opening in black patches the size of the head of a pin or smaller. They are usually numerous and the stem dies rapidly.

The growth of the stem is indeterminate, but as each succeeding section is a little smaller than the one below, the time arrives in the history of each when no more can be pushed out and the growth ceases. The undeveloped internodes soon die and thus the stem, if it grows at all, must put its energy into branches, as the silex coating prevents its increase in diameter.

9. *Robustum* (A. Br.) *E. robustum* A. Br. Stems 3 to 6 feet tall, 2 to 6 lines wide, 16 to 48 angled, simple or branched the second year; ridges rough with cross-bands of silex; grooves naked with a smooth coat of silex, and when young with a thin white scurfy coat that soon falls off; sheaths tight to the stem, or recurved and deciduous in fragments in age, as broad as long, soon developing a black girdle at base, an ashy or pinkish one through the middle and a black one above, the last usually very small, all variable in breadth and intensity of color; leaves linear, sharply 3 angled; commissural groove not widened above; teeth more or less persistent for a season, seldom torn off by the growth of the stem, articulated to the leaves,

cohering, in groups, brown centrally, with tawny margins $\frac{1}{2}$ their height, ending in filiform usually flexuous appendages, the edges beset with unicellular bristles; branches variable in number and length, the sheaths mostly like those of the stem except the teeth always persist and the leaves are usually grooved centrally; spikes usually green, oval, up to an inch long and half as wide, sharply apiculate. *Ramosum* and *polystachyum* forms occur in this as well as in *affine*. 113

Rare east of the Mississippi, where it is replaced by var. *affine*. Very common west, where it has been reported from nearly every State. I have seen it from but six localities in the Eastern States, Wallingford, Pa., *T. C. Palmer*; Towson, Md., *C. E. Waters*; Peoria, Ill., *F. E. McDonald*; Illinois, without locality, *Dr. Brendell*; Mattsville, Ind., *Guy Wilson*; Sarnia, Mich., *C. K. Dodge*; accredited to New Jersey by Milde, and also found in the Himalayas.

Var. *minus* Eng. is simply the same thing reduced, often growing with it. As there is already a variety *minus* of *hiemale* this name will not stand, and the form is of too little moment to merit another.

Stems of this can usually be recognized at a glance, but it is hard to embody the description in words that will enable one to separate it from *affine* at once. From *Californicum* it can only be separated by use of a lens, as their appearance is identical.

10. *Californicum* Milde. Plants of various appearance, now 15 inches high and 4 lines wide, now $7\frac{1}{2}$ to 8 feet tall and 8 lines wide, 25-40 angled; the ridges with two distinct rows of tubercles or occasionally with transverse bands of silex, the grooves abundantly supplied with rosulæ, either in regular rows or scattered, often indistinct on old stems because of a heavy deposit of silex; sheaths as broad as long, with a broad or narrow black or dark brown ring just above the base, an ashy band in the middle and another usually narrow dark band at top. In young plants the sheaths are usually concolorous with the stem save for the terminal band; leaves linear, 3-angled with two rows of tubercles on the middle angle; commissural groove narrow, slightly or not at all widened above; teeth persistent, dark brown, firm, united two-thirds their height by brown borders; or brown-centered, flexuous, membranous-bordered, united or free, or early 114

deciduous, leaving only a small dark brown spot at the tip of the leaves; branches none or few, short or up to 18 inches long, fruited or not, on the upper part of old stems.

Type. California *Balfour*, 1854. I have seen it from the following localities: California: Sacramento, *Wilkes Exp.* (Sheaths black, teeth persistent, near var. *Javanicum*); Berkeley, *W. C. Blasdale* (very stout, often with two rows of stomata); San Rafael, *Munson & Hopkins* (like last, but with one row of stomata). Arizona: Cedar Ranch, *MacDougal*. Nevada: Humboldt Mts., *Watson*. Utah: Fish Lake, *Jones*; Glenwood, *Ward*. Idaho: Peter Creek, *Sandberg*; Salmon, *Henderson*. Oregon: Port Discovery, *Wilkes Exp.* Washington: Tacoma, *Flett*; Klickitat Co., *Suksdorf*. British Columbia: New Westminster, *A. J. Hill*. (No rosulæ, occasionally two rows of stomata, extraordinarily thick coating of silex.)

Except the Berkeley and San Rafael plants these can be told from *robustum* only by aid of the microscope to see the tubercles and rosulæ. Though specimens vary considerably in appearance, the presence or absence of teeth, the size and intensity of the rings, a parallel can usually be found in a good series of *robustum*.

11. *Doelli*. Stems 1½ to 2½ feet high, erect, dark green, 10 to 20 angled, the ridges with two rows of tubercles or short crossbands, the former predominating; grooves with irregular rows of rosettes; sheaths entirely black or with a narrow ashy band which is broader the second year; the leaves plainly 4 angled through the grooving of the central ridge; teeth persistent or becoming broken in age, rigid, erect, dark brown or black, grooved in the center, with narrow white margins and usually deciduous filiform tips. Somewhat resembles a robust *E. trachyodon*, which it is quite near.

Type European. British Columbia, near Wharnock Station, *A. J. Hill*; Vancouver, *Macoun* (as *ramosissimum*); Blacktail Deer Creek, Yellowstone Park, *Knowlton*. The latter is quite peculiar in appearance and approaches *robustum*. None of the specimens exactly agree, but will come here better than elsewhere. *The Ames Botanic Laboratory, North Easton, Mass.*

THE SPECIES-CONCEPTION AMONG THE TERNATE BOTRYCHIUMS.

BY WILLARD N. CLUTE.

Living as I do in the midst of a region rich in specimens of the ternate *Botrychiums*, I have taken more than ordinary interest in the discussion of the relative rank to which the various forms should be assigned. After considerable study of the subject which has consisted of a careful balancing of the degree of differentiation in each form, as well as an examination of much material both in the herbarium and in the field, I have come to certain conclusions which I purpose to set down here.

Before the separate forms are discussed it may be well to say a few words on the variations of *Botrychium ternatum* in general. It is a noticeable fact that all the so-called new species of this section of the genus, have been based primarily upon the cutting of the sterile part of the frond. This is all the more remarkable since there are probably no other genera in which species are founded on the minor outlines of a mere leaf. One has but to turn to nature in any clime to see that leaves are not invariably of the same shape. Note the wide variation in the moonseed, the hollyhock, the sassafras, and some of the buttercups among flowering plants, and if it be contended that the cases are not parallel, take as further illustration the blood-root, which, like the *Botrychium*, produces but one leaf a year, and note the cutting of its single leaf. If all these forms of *Botrychium* are species, why have not the forms of the bloodroot been segregated? Moreover, if we are to recognize these forms of *Botrychium* as species, why should we not also recognize as such the three hundred forms of *Athyrium filix-foemina*, or the hundred or more forms of *Scolopendrium*? It is unavailing to say that these latter are mere gardeners' varieties, for we have it on the authority of Mr. Druery, who is familiar with them all, that a large number come true from spores.

Experiments with flowering plants have shown that the thickness of leaves and the amount of cutting of their edges, may be altered by different

degrees of moisture, sunshine, etc., to which they are exposed, and we may infer as much for the ferns. This being so, it is not difficult to account 116 for the slight variations in cutting exhibited in plants from widely separated points in the United States.

It is, of course, possible to follow the latest writer on the subject, and consider each extreme of variation a distinct species, but I do not agree with him in the opinion that the naming of varieties is a stupid practice, nor do I see that it necessarily follows that because a species was named *Japonicum* from Japanese specimens that we must infer that its centre of distribution is in Japan. As I understand it, to take a familiar example, *B. ternatum* stands for a plant possessing certain characters no matter where found. If we should find another *Botrychium* that differed from this in some specific way, it would be correct to call it another species; but if it showed minor differences, slightly thicker or thinner leaves, a longer or shorter stipe, a little deeper notching of the leaves, etc.—all characters that vary with the locality—then it would seem more properly referred as a variety of the first species.

As I have noted in this journal there are certain slight differences between the Japanese *B. ternatum* and our familiar species of Eastern America, but these are not enough, I now believe, to make them two separate species, since all the differences are found in the texture and cutting of the sterile part of the frond. Under such circumstances I would arrange our American forms as follows:

BOTRYCHIUM TERNATUM OBLIQUUM (*B. obliquum* Muhl.). The common form in the North Atlantic States.

B. T. OBLIQUUM forma DISSECTUM (*B. dissectum* Spreng.). An exact duplicate of the preceding form in everything except the cutting of the pinnules. These latter characterized by a paucity of tissue between the terminal veins. Has the same habitat and range, and the same peculiarity of waiting until July or later before putting up its leaf for the season. No more entitled to specific rank than the “cut leaved” birch or elder.

B. T. OBLIQUUM forma INTERMEDIUM (*B. obliquum intermedium* Unde.). I would call this a mere form, comparable to any of the chance varieties of

Athyrium filix-foemina.

B. T. OBLIQUUM forma COULTERI (*B. Coulteri* Unde.). A western form rather more fleshy than that of the East. Grows in geyser formations which may account for the difference in its appearance. 117

B. T. OBLIQUUM forma OCCIDENTALE (*B. occidentale* Unde.). Closely related to the preceding, and, in my opinion, a phase of it. Both good representatives of the western form.

B. TERNATUM ONEIDENSE (*B. ternatum* var. *Oneidense* Gilbert). This, the most strongly marked of the forms in the Atlantic States failed to receive a place in the recently published index to the described species of *Botrychium*. It can be distinguished at a glance in field or herbarium by its broad and slightly divided pinnules, and is very common in central New York. The fronds, notwithstanding their broad pinnules, are among the smallest of the group. If any of our forms of *Botrychium* are entitled to sub-specific rank, this is certainly the one.

B. TERNATUM SILAIFOLIUM (*B. silaifolium* Presl.). This is also mainly a western form. To it, however, I would refer the plant recently described by Mr. Gilbert as *B. obliquum Habereri* from central New York. I have examined the type specimen and in my opinion it agrees perfectly with specimens of *silaifolium* from California identified by Dr. Underwood. Even the striations produced in the pinnae by drying appear identical. I should call this a sub-species, as it does not approach the type as closely as the others.

So little is known about *B. biternatum* Unde. and *B. tenuifolium* Unde. that I shall not attempt to place them. Judging from what I have seen of the latter, and I have seen numerous plants in the field, I should consider it a form of *obliquum* and I suspect that *biternatum* will prove to be based upon aberrant plants of this which have fruited in spring instead of autumn. In regard to this, Mr. W. W. Ashe has recently informed me that many spring-flowering southern plants do not flower in the North until late summer. It is possible our ferns may have similar changes in their fruiting season.

NEW FORMS OF FERNS.

BY CHARLES T. DRUERY, F.L.S.

The editorial note appended to my short article in the April issue of THE BULLETIN rather takes my breath away, as I never imagined that an answer to my query could “depend somewhat upon whether we admire ferns for pure leaves or whether we collect them for study.” No true fern lover in either case would knowingly destroy one of Nature’s own novelties in the way I described by denuding it repeatedly of its fronds for herbarium purposes in situ, when by removing and cultivating it he could also, in either case, not only gratify his own special taste more fully, but could afford much gratification to a host of other fern lovers of either class. That “students of ferns know that many fern forms are due to varying conditions of soil, light, moisture, etc., and are inclined to pay very little attention to them” I accept at once, ranking myself with them; but advanced students also know that many forms are not demonstrably due to such influences, and amongst these forms are all those which claim so much attention in this side of the ocean. The former are, as the editor puts it, “variants,” the latter true varieties, and so far as they are of Nature’s own shaping, i. e., wild finds, as distinct from improved selected types from the spores, they have at least as much right to recognition in fern literature as the normals. Hence it is to be regretted that a unique form of the Christmas fern (*Polystichum acrostichoides*) should exist in the possession of a member of the Fern Chapter for ten years, and, yet, never be described. What have the other members done that such interesting data to some of them should be withheld?

The reference to seven-toed kittens and two-headed rabbits, as fair parallels to the finest fern varieties in the mind of the average student, is a poor compliment to the student who would certainly benefit by a better acquaintance with the plumose section of varieties at any rate. With the many botanists stated to exist in the States who “prefer a wild rose to all the gardeners’ many-petalled creations” I have more sympathy, but here comes in the old botanical mistake embodied in the term “garden forms” of ferns

as applied to all varieties, with the wild finds of which the gardeners have had nothing whatever to do. What would such a botanist do if in his rambles among the wild roses he came across a Marechal Niel as a wild sport? That is a fair parallel to some of our best wild finds as compared with the normal types, and he would be a singular man, I opine, in more senses than one, if he turned up his nose at it as a mere variant and held his tongue for ten years without describing it. I am gratified to the editor for holding all his abnormal specimens at my disposal, but, reading between the lines, I fear they would embrace no acquisitions from my point of view, or he would not be so ready to part with them. I hope sooner or later he will come across a thoroughbred and become thereby a convert to my theory, that constant and symmetrical variations are fully as much, if not more, entitled to both lay and scientific attention than the normal specific forms from which, by some occult process, they arise under natural conditions.

119

[It is doubtless as difficult for Mr. Druery to understand our position in this matter as it is to understand his. How a *student* of ferns can care for what might be termed abnormal variations is beyond our comprehension. The student is always interested in normal variations, if we may so describe the common, slight variations in form and texture due principally to ecological factors; in fact, it is necessary that we take all such into account in order to get a correct average of the species; but to give serious attention to forked, crested, plumed, tasselled and befrizzled specimens of ferns, which are manifestly due to the slipping of a cog somewhere in Nature's machinery, is quite out of the question. We grant that some of these attain forms that merit admiration for their beauty, or oddity, as showing what Nature can do in the way of leaves, but we maintain that were these forms animal, instead of vegetable, they would excite only feelings of repulsion. Now, the student of fern species is quite inclined to think of these "freaks," as he calls them, much as others would if they *were* animal. The botanist may admire the form, hue and perfume of the gardener's rose, but this is not the rose he cares to study. In the early numbers of THE FERN BULLETIN, upward of sixty American ferns have been put on record as bearing forked or tasselled fronds, and so far as the editor is aware, not one of these has been taken into cultivation and only a very few have been given names. This fact will probably explain our position to some extent. When the editor has leisure, he is going to dig up every one of these variants in his own

120

locality and send them to Mr. Druery, in anticipation of which it would be well for the latter to consult his gardener and glazier about an extension to his ferneries.—ED.]

FERNS IN BOTTLES.

A correspondent sends us the following clipping from the *Westminster Gazette*. We are unable to vouch for its accuracy, but as it may give some cultivator a hint we reprint it in full.

In a beautiful garden at Crouch End, belonging to one of the few old world bowers which have withstood the tempting offers of the building speculator, may be seen one of the queerest freaks that Nature has ever played in park or garden. About three years ago a long row of glass ginger bottles were placed neck downward in the ground, with a few inches of the other end projecting to form a border for the kitchen garden paths. Each of these bottles now contains a fairy-like resident in the shape of a dainty little fern, perfect in form and color, and of many varieties, the ribbon fern and hart's-tongue predominating. As no ferns had at any time been planted in that part of the garden it is amazing how they got there. Perhaps Nature thought it foolish to waste so many little natural hothouses, and put in each a pinch of the stuff she makes ferns of. If so, she must view with much pride the result of her experiment.

Miss Angie M. Ryon, Niantic, Conn., reports finding fine plants of *Ophioglossum vulgatum* upon a very rocky hillside, the roots crowding themselves between the bits of rock that had been broken up by loads of heavy timber passing over them the previous year. The plants were exposed to the full rays of the sun for most of the day.

WILLIAM RALPH MAXON.

William Ralph Maxon, whose portrait is presented this month, first saw the light at Oneida, N.Y., on Feb. 27, 1877, where his parents reside. He graduated at Oneida High School in the class of 1894. From there he went to Syracuse University, where he took the degree of Ph. B. in 1898. The bent of his mind was toward botany and almost immediately after graduating he went to New York and was employed for a few months in the herbarium of the Botanical Garden at Bronx Park. From there he went to Washington and took a temporary position in the U.S. National Museum. But in August, 1899, as the result of a Civil Service examination, he received the appointment of Aid in Cryptogamic Botany in that institution, and still retains that position.

Mr. Maxon joined the Fern Chapter in 1895, served as its secretary for the year 1899, and as president of the Chapter for the two years 1900 and 1901. He is a member of the Botanical Society of Washington, of the American Association for the Advancement of Science, and of two other local societies in Washington. He has published several valuable papers on botanical subjects, among which are "A List of the Ferns and Fern Allies of North America north of Mexico;" "On the Occurrence of the Hart's-tongue in America," which formed his presidential address before the Fern Chapter at its meeting in New York in 1900; "A Study of Certain Mexican and Guatemalan Species of Polypodium"; besides several shorter papers containing descriptions of new species, including a continued series in the FERN BULLETIN entitled "Notes on American Ferns." For a young man Mr. Maxon has done some notably good work, and his position in the National Museum is one which will give him exceptional opportunities for original work in the future.—*B. D. Gilbert.*

ANOTHER STATION FOR ASPLENIUM EBENEUM HORTONAE.

It affords me much pleasure to report another station for the remarkable fern now known as *Asplenium ebeneum Hortonae*.

A single fine specimen some eight or ten inches high was discovered by Miss K. A. French at the base of a ledge in Pittsford, Vt. No sign of fertility is discernible in the old fronds decaying about the base, the season's growth or the newer fronds stretching up indoors. Whence came this beautiful variation in the midst of type specimens galore?—G. A. Woolson, *Pittsford Mills, Vt.*

INDEX TO CURRENT LITERATURE RELATING TO FERNS.

Readers are requested to call our attention to any omissions from this list.

CLUTE, W. N. *Fernwort Notes—III*. Fern Bulletin, Jl. 1903.

CHRIST, H. *Can Scolopendrium Lindenii be Separated From S. Vulgare?* Fern Bulletin, Jl. 1903.

EATON, A. A. *The Genus Equisetum in North America*. Fourteenth Paper. Fern Bulletin, Jl. 1903.

FITZPATRICK, F. J. and M. F. L. *The Fern Flora of Iowa*. Fern Bulletin, Jl. 1903.

FLETT, J. B. *The Fern Flora of Washington*. Fern Bulletin, Jl. 1903.

GILBERT, B. D. *Asplenium muticum*. Fern Bulletin, Jl. 1903.

GILBERT, B. D. *Two New Varieties of the Ternate Botrychium*. Fern Bulletin, Jl. 1903.

HAESSELBARTH, F. C. *The Walking Fern*. American Botanist, Jl. 1903.

ORCUTT, C. R. *Ferns of Southern California*. West American Scientist, Ag. 1903.

ORCUTT, C. R. *Isoetes of Southern California*. West American Scientist. S. 1903.

SHULL, C. H. *Geographic Distribution of Isoetes Saccharata*. Botanical Gazette, S. 1903.

STILLMAN, B. W. *The Climbing Fern*. American Botanist, Je. 1903.

EDITORIAL.

The editor of this journal is again away from home, which will explain any delays in his correspondence. Letters intended for him and sent to the usual address will be forwarded at once.



Every time the FERN BULLETIN is late, doubtless half the subscribers wonder what the editor can be doing to prevent the magazine from appearing on time. The editor, therefore, rises to explain that this publication would always be issued promptly if correspondents would send in copy in season. We endeavor to have each issue ready for the printer *a month before publication*, and yet there are many of our contributors, who, having arranged for space in a number, will wait until two or three days before the publication date to send in copy. There is too much matter in the FERN BULLETIN for it to be printed, bound and mailed in one day—or in one week for that matter—and we simply cannot appear on time if we have to wait for copy. We trust that contributors will bear this in mind and co-operate with us in future in this matter of promptness.



In this number, all subscribers whose subscriptions are not paid in advance, will find a bill for the next volume, and the amount in arrears, if any. There are so few who stop their subscriptions after once becoming members of the FERN BULLETIN circle of readers that to save all from the possible loss of a number between the expiration and renewal of subscription, we continue to send the magazine until ordered to stop. Those who do not wish to be considered subscribers for the new volume should notify us at once. In view of the liberal terms upon which the publication is sent, we trust that we may continue to count all our present subscribers among our readers for another year.



The new volume of this magazine will be along the lines of its predecessors. The Fern-floras of the States will be continued, those for California and Florida being expected to appear early in the year, 124 followed by others now in preparation. The series of portraits of fern students will be discontinued for the present, to make room for a new series on exotic ferns in which will be illustrated the ferns of unusual appearance from other lands, together with descriptive notes upon their haunts and habits. In nearly every fern collector's herbarium are species of which little more than the name is known. It is expected that this series will add much to our knowledge of these. Mr. Eaton's *Equisetum* articles will be finished during the year. Other features of the publication will be continued. A cordial invitation is extended to every reader to contribute notes and articles of interest.



When one's herbarium has grown so bulky that it cannot be looked through in an hour or so, and especially when it contains numerous sheets of the same species, it often becomes a puzzling question what to show the non-scientific visitor who wants to see the ferns, but who has no interest in the slight differences that separate closely related forms. To fit such cases we would suggest the formation of an "oh, my!" collection—a collection designed to provoke the visitor's interest and admiration and draw forth frequent ejaculations of surprise. Such a collection saves wear and tear on the general herbarium and often excites a real and lasting interest in the ferns. It should contain, of course, the walking and climbing ferns, the little *Schizaea*, the hartstongue, the maidenhair, the cinnamon and sensitive ferns, etc. Some of the gold and silver ferns might be included, the star fern is desirable and various species of grape fern will add to the interest. A few finely cut fronds like that of *Dicksonia* may be added, with such other species as suggest themselves on account of oddity in fruiting. A few fern allies would make a complete and desirable show herbarium.



The recent likening, in this journal, of crested and tasseled fronds to two-headed rabbits has borne fruit in an unexpected quarter. Certain cultivators

of ferns now speak of their stock as species, varieties and two-headed rabbit sorts!



In a recent article on the distribution of *Isoetes saccharata*, in the *Botanical Gazette*, the author notes that *I. saccharata* is found only in Chesapeake Bay, while *I. riparia*, a form that closely resembles it, is found only in Delaware Bay, and that, notwithstanding the small differences between the two, there have been described two intermediate forms, *I. saccharata Palmeri* and *I. s. reticulata*. From this and other facts, he inclines to the belief that *I. riparia* is simply an extreme form of *saccharata*. A curious circumstance connected with the species and forms is that specimens collected in a certain locality appeared one season as the type and the next as the variety. This is not the only instance on record of *Isoetes* species intergrading. Mr. A. A. Eaton has noted that the variety *Californica* of *I. melanopoda* intergrades on the one hand with *I. Howelli* and on the other with the type, while specimens referred to *I. mexicana* by Underwood have since been identified as *I. Orcuttii* and *I. melanopoda*. All this seems to indicate that the systematists have not yet got hold of the proper characters upon which to found specific differences, and there is here a chance for a philosophical botanist to distinguish himself. It is not enough that the plants look different; the differences must be specific. All of the forms of *Equisetum arvense* look different, but they are not species by any means.



Plans are being made for another meeting of fern students at St. Louis this winter, during the session of the American Association for the Advancement of Science. There is a prospect of some excellent papers to be presented, and a large and enthusiastic meeting is assured. Programs and other information may be obtained of Prof. N. L. T. Nelson, Central High School, St. Louis, Mo., or of the editor of THE FERN BULLETIN. The editor expects to be at the meeting and hopes to meet there the majority of his readers.

BOOK NEWS.

With two books like “Our Ferns in Their Haunts” and “How to Know the Ferns” in the field, some might think that the ground is too thoroughly covered to leave room for another, but Dr. Waters has proven 126

^[1] otherwise in his new volume with the simple title of “Ferns.” If the older books are taken as books designed for beginners, the new one may be described as one step more technical—a connecting link between popular handbook and scientific manual. In the matter of information it contains practically nothing that has not been published before, being primarily a rearrangement of what is known of our ferns and bearing internal evidence that the pages of the popular works above mentioned have supplied much help in its making. This is especially noticeable in the appropriation of original stanzas from “Our Ferns in Their Haunts,” without giving credit for them. Even when giving credit the author is not always fortunate, as when he ascribes to Miss Pratt certain lines that are Campbell’s. The book is written from the standpoint of the author’s experiences in the vicinity of Baltimore, and as such cannot always be taken as representative of ferns and fern habitats in other parts of our country. The book is also remarkable for the entire absence of author citations for the generic and specific names—in this being unique among American fern books. This will make it difficult for beginners who use this book to look up the species elsewhere. The entire subject of the authority for the names is left untouched in the part that explains why scientific names are used. There is also evident a disinclination to give credit for recent work, the usual statement being that such and such varieties “have been described,” without noting where and by whom. Since these descriptions were heretofore to be found mostly in periodicals, the desirability of mentioning their place of description is apparent. This much being said in criticism of the book, there is yet much to praise. The illustrations are well executed and include a fair proportion of views in the habitats of the ferns, while the photographs of the sori enlarged are exceptionally valuable. Anything like them have never before been published. The analytical key based on the stipes has been well and carefully worked out. It forms a very instructive chapter, though such a key

will rarely be consulted by the beginner, because the usual keys are easier. The book is also of interest for the number of varieties or forms 127 included, many of which are not mentioned in the manuals, and for numerous photographs illustrating these forms. There is also a chapter on fern photography. The nomenclature is conservative; old and well known names having been retained in spite of recently proposed substitutes. For his stand in this matter the author is to be greatly commended. The book can in no sense be considered a rival of others in the field. It fills a place of its own and as such will be hailed by the fern loving public as another aid to the proper understanding of this most beautiful and attractive section of the vegetable kingdom. Type, paper and presswork all combine to make this a handsome as well as useful book.

[2]

In the making of his “Flora of the Southeastern United States” Dr. Small has practically worked over anew all the species of that region and the results of this stupendous undertaking are now presented in a bulky octavo volume of nearly fourteen hundred pages. Dr. Chapman’s flora of the same region is fairly conservative, while this one goes to the opposite extreme, being noticeable for the greatly increased number of species and an equal lack of what the older book is inclined to consider varieties or forms. This gives students a choice of books, but it is probable that a beginner will have much difficulty in identifying his plants by the aid of the new one, because of the many closely related forms described as separate species. The book will appeal more to advanced students specializing in a few orders or genera. The nomenclature is, of course, the most radical, even extending beyond the genera and species to the orders, so that new names are proposed for the Leguminosæ, Labiataë, Scrophulariaceæ and many others. Along with the segregation of species there has been a similar splitting of genera. To the individual familiar with the usual manuals the book will appear almost like the flora of a foreign land. The author, however, has been painstaking and conscientious throughout the work, adhering closely to the ideals prevalent at New York and a few other centers of botanical activity, and if he has produced a volume that will become the object of much 128 criticism he has also produced an excellent illustration of what the recently proposed rules of nomenclature, and new conceptions of species are capable of when allowed full swing. Although not agreeing with his conclusions, even conservative botanists will appreciate the author’s

consistency and he is to be congratulated upon the completion of his work. To those who prefer the less elaborate manual, Dr. Small's book will always be indispensable for reference.

Mr. Charles T. Druery, well known on this side of the Atlantic as an enthusiastic cultivator of ferns, has issued a book devoted principally to British fern varieties, which he has named "The Book of British Ferns."^[3] In this work he has been assisted by various members of the British Pteridological Society, an association similar to our Fern Chapter, and of which Mr. Druery is president. As is well known the British fern hunter and fern cultivator is most interested in the abnormal forms of ferns, and as a result so many of these have been described that a complete list numbers more than a thousand, though there were less than seventy-five native species to begin with. Mr. Druery's task has been to select from this list the really meritorious varieties from the cultural standpoint. To this revised list he has added papers on the culture and propagation of ferns, fern hunting, fern crossing, apospory and kindred subjects, making a volume which should be of interest to American readers for the light thrown upon fern life. The book is well printed and well illustrated, the forms shown being mostly the more striking varieties.

A WORD FROM THE EDITOR

For more than two years the editor of the FERN BULLETIN has also been editing a journal for the plant-lover, called THE AMERICAN BOTANIST. That he has been fairly successful may be assumed from the fact that it now has the largest circulation of any botanical magazine in America. People do not buy such publications out of charity; they buy them for what they contain. No doubt the principal reason for the AMERICAN BOTANIST's popularity is that it is untechnical—even those who are not botanists can understand it. Moreover it deals with a very different side of botany from that usually presented. If you are interested in plants as living things—their uses, habits, and curious methods of getting on in the world—this is just the publication you want. A large number of fern students are already readers of the AMERICAN BOTANIST, but to induce others to become such, we offer the last three numbers for this year, all the numbers of 1904, for the regular subscription price of \$1.00 if received before the first of January. Or we will send the first five volumes (of six numbers each) and a year's subscription, for \$3.00. With the latter offer, your subscription to the FERN BULLETIN will be renewed for 50 cents additional.

Address WILLARD N. CLUTE, BINGHAMTON, N. Y.

For Christmas

Nearly every fern student seems to have a copy of "Our Ferns in Their Haunts," but if you happen to know of one who doesn't, you could scarcely do a more graceful thing than to give him one for Christmas. You may be sure the book will be consulted many times next year and in the years to follow and every time this happens the giver will be thought of with pleasure. If you have a young friend who is beginning to get interested in Nature, crystalize his tendency by giving him this book. The illustrations will make him a lover of ferns and the text will make him wise about them. There are 225 illustrations and 340 pages of text. No other fern book is so full, so clear or more accurate. The key for identifying the ferns has illustrations of fruit-dots and even a child can name the ferns by its use. Sent postpaid upon receipt of \$2.15. Address

Willard N. Clute & Co., Binghamton, N. Y.

FOOTNOTES

- [1] Ferns. By Campbell E. Waters. New York. Henry Holt & Co. 1903. 8 vo. 360 pp. \$2.75 net.
- [2] The Flora of the Southeastern United States. By John K. Small. New York. Published by the author. 1903. 8 vo. 1,375 pp. \$3.60 net.
- [3] The Book of British Ferns. By Charles T. Druery, F. L. S., V. M. H. London. George Newnes, Ltd. 1903. 12 mo. 135 pp. \$1.25.

SEND AT ONCE FOR MY
JUNE, 1903, SPECIAL BULLETIN OF
Microscopes and Objectives Cameras,
Photographic Lenses, Etc.



Bear in mind I guarantee every instrument to be as represented and in good working order and adjustment. Prices in parenthesis indicate list price, by which is understood present or latest price. This stock is constantly changing; therefore, if interested, place order at once. Send name and address if you would like to receive these lists as issued.

A SPECIAL OFFER

To facilitate the purchase of such a microscope as shall with certainty prove satisfactory and adequate to its intended use, I am willing to make the following proposition whereby YOU RUN NO RISK of getting an unsuitable instrument. The offer is: Order what microscope you like from this list; use it for one or two months; if then you decide that you want another one on my list, or a new one of

whatever make, order it from me and return the other one (prepaying expressage) and I will allow you FULL PRICE PAID.

EDWARD PENNOCK
DEALER IN OPTICAL INSTRUMENTS
3609 Woodland Ave.
Opposite Medical Department, University of Pennsylvania
PHILADELPHIA

FOR SALE

We offer a set of the FERN BULLETIN, lacking only two numbers of Vol. 5, two numbers of Vol. 4 and four numbers of Vol. 1-3, for the very low price of \$6.00 postpaid. This set includes 36 of the 44 numbers thus far issued. Send order by postal. First order secures the set. Address

FERN BULLETIN. BINGHAMTON, N. Y.

Transcriber's Notes

- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- Silently corrected a few palpable typos.
- Generated a spine image based on elements in the cover.
- Added a Table of Contents.
- In the text versions only, text in italics is delimited by underscores.

*** END OF THE PROJECT GUTENBERG EBOOK THE FERN
BULLETIN, OCTOBER 1903 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.